QUESTION 1

THE FUNDAMENTAL EQUATION OF THE SOLOW GROWTH MODEL

1. Basic Setup of the Solow Model

The economy produces a single good using capital (K), labour (L), and technology (A).
The aggregate production function is assumed to have constant returns to scale:

$Y = F(K, A\ L)$

Technology is labour-augmenting. We convert the model to intensive form (rewriting the model in terms of variables per unit of effective labour) by dividing by effective labour AL.

Define:

$r = K / (A\ L)$ (capital per effective labour)
$y = Y / (A\ L) = f(r)$

The function f(r) is increasing and concave.

Assumptions:

- A constant fraction s of output is saved: $S = sY$
- Investment = saving
- Capital accumulation: $\dot{K} = sY – \delta K$
- Labour grows at rate n
- Technology grows at rate a
- Depreciation rate is δ

Goal is to derive a dynamic equation for r(t).

By definition:

$r = K / (A\ L)$

Rewriting as:

$K = r\ A\ L$

Differentiating both sides with respect to time:

$\dot{K} = d/dt\ (r\ A\ L)$

Using product rule :

Differentiating:

$dK/dt = d(r\ A\ L)/dt = \dot{r}\ A\ L + r\ \dot{A}\ L + r\ A\ \dot{L}$.

Now dividing the whole expression by A L:

$\dot{K} / (A\ L) = (\dot{A}/A)\ r + (\dot{L}/L)\ r + \dot{r}$

Rearranging to isolate $\dot{r}$:

$\dot{r} = \dot{K}/(A\ L) – (\dot{A}/A + \dot{L}/L)\ r$

Putting the growth rates of A and L as what we denoted:

$\dot{A}/A = a$
$\dot{L}/L = n$

Substituting:

$\dot{r} = \dot{K}/(A\,L) - (a + n)\,r$

From capital accumulation:

$\dot{K} = sY - \delta K$

Substitutin this into the equation:

$\dot{r} = (sY - \delta K)/(A\,L) - (a + n)\,r$

Now dividing Y by AL and K by AL:

$Y/(AL) = f(r)$
$K/(AL) = r$

Substituting these:

$\dot{r} = s\,f(r) - \delta\,r - (a + n)\,r$

Combine terms:

$\dot{r} = s\,f(r) - (n + a + \delta)\,r$

This is the fundamental differential equation of the Solow model:

Fundamental Equation:
$\dot{r} = s\,f(r) - (n + a + \delta)\,r$

3. Cobb–Douglas Special Case

Let the production function be: (*α (alpha) is the output elasticity of capital, α tells you what fraction of output comes from capital.)

$Y = K^{\alpha}\,(A\,L)^{(1-\alpha)}$

Dividing by AL:

$y = f(r) = r^{\alpha}$

Substituting $f(r) = r^{\alpha}$ into the fundamental equation:

$\dot{r} = s\,r^{\alpha} - (n + a + \delta)\,r$

This is the standard Cobb–Douglas Solow equation.

4. Steady State

At steady state $\dot{r} = 0$:

$s\,f(r^*) = (n + a + \delta)\,r^*$

For Cobb–Douglas:

$s (r^*)^\alpha = (n + a + \delta) r^*$

$r^* = [s / (n + a + \delta)]^{(1 / (1 – \alpha))}$

then, steady-state output per effective labour is:

$y^* = f(r^*) = (r^*)^\alpha$

5. Economic Intuition

The fundamental equation separates two forces:

1. **Investment per effective worker: s f(r)**
   Higher r produces more output, and a fraction s becomes investment.
   This term increases capital per effective labour.

2. **Capital dilution and decay: (n + a + δ) r**
   • Population growth n spreads capital over more workers
   • Technological progress a increases effective labour AL, diluting capital
   • Depreciation δ wears out capital

Net accumulation is:

Change in r = (investment) – (dilution and depreciation)

If investment > dilution, r rises.
If investment < dilution, r falls.
This pushes the system toward a unique steady state.

Saving affects the level of income, not long-run growth.
Long-run per-worker growth is driven by a (technology).

The Solow model is used for:

• Growth accounting (decomposing growth into capital, labour, and technology)
• Predicting convergence of poorer countries under similar parameters
• Policy evaluation: how saving, population policies, and education affect levels
• Understanding why productivity (A) matters for long-run standards of living
• Calibrating growth simulations for central banks and planning agencies

Before Solow, the Harrod–Domar model predicted "knife-edge" instability because the capital-output ratio was fixed. Solow (1956) introduced substitutability between capital and labour using a neoclassical production function. This made the economy stable and allowed the capital-output ratio to adjust endogenously. The innovation fixed Harrod–Domar instability and clearly separated:

• level effects -- saving and capital accumulation
• growth effects --- technology

Why the Solow Model Is a Supply-Side Model

The model is considered supply-side because:

• Output is determined by the availability of factors (K, L, A), not by demand
• Prices and markets are competitive; there is no unemployment or demand deficiency
• Saving determines investment; there is no role for aggregate demand or government spending multipliers
• Technology and factor accumulation fully determine long-run income
• Long-run growth in per-worker output is driven entirely by supply-side productivity (A)

In the Solow model, demand plays no role.
Everything important comes from production capacity and technology.

QUESTION 2:

TRANSFORMING THE FIRST-ORDER NONLINEAR EQUATION INTO A LINEAR ONE

In the Cobb–Douglas Solow model, the fundamental differential equation is:

$\dot{r} = s\, r^{\alpha} - (n + a + \delta)\, r$

This is a nonlinear first-order ODE because of the term r^{α}.

To solve it analytically, we use a Bernoulli transformation.

Rewriting the equation
Move all r terms to one side:

$\dot{r} + (n + a + \delta)\, r = s\, r^{\alpha}$

Recognising it as a Bernoulli equation
A Bernoulli equation has the form:

$\dot{x} + p(t)\, x = q(t)\, x^{m}$

Here:
$x = r$
$p(t) = n + a + \delta$ (a constant)
$q(t) = s$ (constant)
$m = \alpha$

Since $\alpha \neq 1$, it matches the Bernoulli form.

Applying the Bernoulli substitution

Let $r(t)$ be capital per effective worker and let α be a constant with $0 < \alpha < 1$. Defining the new variable $z(t) = r(t)^{(1 - \alpha)}$.

Differentiating z using the chain rule

Writing z as $z = r^{(1-\alpha)}$. Differentiating with respect to t.

$dz/dt = d/dt\, [\, r^{(1-\alpha)} \,]$

Using the power rule and chain rule: $d/dt\ [\ r^p\ ] = p\ r^{(p-1)}\ dr/dt$, where $p = 1 - \alpha$.

So $dz/dt = (1 - \alpha)\ r^{(1-\alpha) - 1} \cdot dr/dt$.

Simplify the exponent $(1 - \alpha) - 1 = -\alpha$. Hence

$dz/dt = p\ r^{-\alpha} \cdot \dot{r}$.

Now substituting $\dot{r}$ from the Solow Cobb–Douglas ODE

The original nonlinear ODE for r is $\dot{r} = s\ r^\alpha - (n + a + \delta)\ r$,

where s, n, a, δ are constants. Replace $\dot{r}$ in the expression for $\dot{z}$:

$\dot{z} = (1 - \alpha)\ r^{-\alpha} \cdot [\ s\ r^\alpha - (n + a + \delta)\ r\ ]$

Distributing $r^{-\alpha}$ inside the brackets term by term

First term product: $r^{-\alpha} \cdot s\ r^\alpha = s \cdot r^{-\alpha + \alpha} = s \cdot r^0 = s$.

Second term product:

$r^{-\alpha} \cdot [\ -(n + a + \delta)\ r\ ] = -(n + a + \delta) \cdot r^{-\alpha} \cdot r = -(n + a + \delta) \cdot r^{-\alpha + 1} = -(n + a + \delta) \cdot r^{1 - \alpha}$.

So after distribution:

$\dot{z} = (1 - \alpha)\ [\ s - (n + a + \delta)\ r^{1 - \alpha}\ ]$

we know $r^{1 - \alpha}$ as z

Replacing $r^{1 - \alpha}$ by z:

$\dot{z} = (1 - \alpha)\ [\ s - (n + a + \delta)\ z\ ]$.

Distributing the factor $(1 - \alpha)$ across the bracket:

$\dot{z} = (1 - \alpha)\ s - (1 - \alpha)\ (n + a + \delta)\ z$.

Bringing the z term to the left-hand side to present the linear standard form:

$\dot{z} + (1 - \alpha)\ (n + a + \delta)\ z = (1 - \alpha)\ s$.

This is a first-order linear ordinary differential equation in z with constant coefficients. The nonlinearity in r has been removed by the substitution $z = r^{1 - \alpha}$.

Algebraically, multiplying $\dot{r}$ by $r^{-\alpha}$ converted the r^α term into a constant s and converted the r term into $r^{1-\alpha}$, which is exactly z. That makes every term either $\dot{z}$, z, or a constant, so the equation is linear in z. Economically, the transformation converts the diminishing-returns nonlinearity r^α into a constant-driving term for the transformed variable and expresses the dilution effect as proportional to z, giving a clear linear decay-plus-constant-drive structure.

ECONOMIC INTUITION

1. The nonlinearity comes from **diminishing returns to capital**
   The term r^α with $\alpha < 1$ captures diminishing marginal productivity of capital.
   This is why the original differential equation is nonlinear.

2. The Bernoulli transformation is a mathematical trick
   It does **not** change the economics.
   It simply changes variables so the diminishing-returns term becomes linear.

3. This helps because a linear ODE has a closed-form solution.
   This allows us to explicitly solve for the time path r(t), which shows:

• how fast the economy converges to the steady state
• how capital per effective worker evolves over time
• how savings, depreciation, technology, and population growth affect the path

4. Economic meaning of the transformation
   By converting the equation into a linear one, we isolate the part of capital accumulation that behaves like regular growth/decay and separate it from the diminishing-returns component.

It turns the messy "curved" dynamics into something tractable, letting us see clearly:

• a constant pull downward: $(n + a + \delta)$
• a constant push upward: s
• the diminishing-returns slowing effect embedded in the exponent $1-\alpha$ of z

The transformation is purely mathematical, but it reveals the clean underlying structure of convergence: investment pushes the economy up, dilution pulls it down, and diminishing returns slow the rate at which the gap to steady state closes.

QUESTION 3

The differential equation of the Solow model in intensive form is:

$$\dot{r} = sr^{\alpha} - (n + a + \delta)\, r$$

A fixed point (equilibrium) satisfies:

$$\dot{r} = 0$$

So solving:

$$sr^{\alpha} - (n + a + \delta)r = 0$$

Factoring out r:

$$r(sr^{\alpha-1} - (n + a + \delta)) = 0$$

This gives two fixed points:

Fixed point 1: $r = 0$

This always exists.

Fixed point 2: Positive steady state r^*

Solve the term in brackets:

$$sr^{\alpha-1} = n + a + \delta$$

Rearranging:

$$r^{\alpha-1} = \frac{n + a + \delta}{s}$$

Raise both sides to the power $1/(\alpha - 1)$(note: α−1 < 0):

$$r^* = (\frac{s}{n + a + \delta})^{1/(1-\alpha)}$$

This is the standard Solow steady state.

The two fixed points of the Solow model are:

1. $\boldsymbol{r = 0}$

(the trivial equilibrium)

2. $\boldsymbol{r^* = (\frac{s}{n+a+\delta})^{1/(1-\alpha)}}$

(the positive, economically meaningful steady state)

Stability of the positive fixed point r* using the substitution method (full homogeneous + particular + general solution)
Define $A = n + a + \delta$ to shorten notation. The ODE is
$\dot{r} = s\, r^{\alpha} - A\, r$
Introduce the substitution $z = r^{(1-\alpha)}$.

After substituting $z = r^{(1-\alpha)}$:

$\dot{z} + (1-\alpha)(n+a+\delta)z = (1-\alpha)s$

Let:

- $L = (1-\alpha)(n+a+\delta)$
- $M = (1-\alpha)s$

The equation becomes:

$\dot{z} + Lz = M$

Solving the homogeneous equation

The homogeneous part is: $\dot{z} + Lz = 0$

Its solution becomes:

$z_h(t) = K e^{(-Lt)}$

where K is a constant determined by the initial condition.

Find a particular solution

Trying a constant solution z_p.
For a constant, $\dot{z}_p = 0$, so substitute into the equation:

$0 + L z_p = M$

Hence:

$z_p = M/L = (1-\alpha)s / [(1-\alpha)(n+a+\delta)] = s / (n+a+\delta)$

General solution for z(t)

Combine homogeneous + particular:

$z(t) = z_p + K e^{(-Lt)}$

Use the initial condition $z(0) = r_o^{(1-\alpha)}$:

$z(0) = z_p + K$

So: $K = r_o^{(1-\alpha)} - z_p$

Therefore:

$z(t) = s/(n+a+\delta) + [r_o^{(1-\alpha)} - s/(n+a+\delta)]\, e^{(-Lt)}$

where $L = (1-\alpha)(n+a+\delta)$

Recovering r(t) from z(t)

We used the substitution $z = r^{(1-\alpha)}$.
Invert this:

$r(t) = [z(t)]^{(1/(1-\alpha))}$

So:

$r(t) = \{s/(n+a+\delta) + [r_0^{(1-\alpha)} - s/(n+a+\delta)] e^{(-Lt)}\}^{(1/(1-\alpha))}$

Long-run value: steady state

As $t \to \infty$, the exponential term goes to zero: $e^{(-Lt)} \to 0$

Thus: $z(t) \to z_p = s/(n+a+\delta)$

Therefore:

*$r = [s/(n+a+\delta)]^{(1/(1-\alpha))}$**

Speed of convergence

The distance between z(t) and the steady state z_p is:

$z(t) - z_p = [r_0^{(1-\alpha)} - z_p] e^{(-Lt)}$

Thus it decays exponentially with rate:

$L = (1-\alpha)(n+a+\delta)$

Deviations satisfy:

$z(t) - z_p \sim e^{(-Lt)}$

The convergence speed is determined by the constant L.

From z(t):
$z(t) - z_p = [r0^{(1-\alpha)} - z_p] e^{(-L t)}$
This decays exponentially with rate $L = (1 - \alpha) A > 0$

Because the mapping r to z is continuous and strictly monotone for $r > 0$, r(t) converges to r* for any positive initial r_0. Therefore r* is locally asymptotically stable and in fact globally attractive on the domain $r > 0$ (every positive initial condition converges monotonically to r*). The convergence rate for small deviations is exponential with rate L. This is the homogeneous-plus-particular calculation proving stability and giving the exact solution.

Linearization check at r* (consistency and local rate)
Defining $g(r) = s r^{\alpha} - A r$.

Computing $g'(r) = s\,\alpha\, r^{(\alpha-1)} - A$.
At $r = r^*$, use equilibrium condition $s\,(r^*)^{(\alpha-1)} = A$ to get
$g'(r^*) = \alpha A - A = -(1-\alpha) A = -L$
So the linearized equation near r^* is $\dot{e} = g'(r^*)\, e = -L\, e$ and $e(t) = e_0\, e^{(-L\, t)}$. This matches the rate L found from the z-solution and confirms r^* is asymptotically stable.

Stability of $r = 0$
Attempting linearization at $r = 0$ requires evaluating $g'(0) = s\,\alpha\, 0^{(\alpha-1)} - A$.

Because $0 < \alpha < 1$, $\alpha - 1 < 0$, and $0^{(\alpha-1)}$ is singular (it diverges to $+\infty$).

So $g'(0)$ is not finite and linearization is not valid at $r = 0$. A different approach is required to determine stability of $r = 0$.

For very small positive r, we compare the two terms $s\, r^{\alpha}$ and $-A\, r$. Since $\alpha < 1$ and $r \to 0^+$, $r^{\alpha} >> r$. Therefore for sufficiently small r we can neglect $-A\, r$ and approximate the ODE by the dominant equation:
$\dot{r} \approx s\, r^{\alpha}$
This separable ODE can be solved.

> We do the integration because the equilibrium at $r = 0$ cannot be studied using linearization.
> The derivative $g'(0)$ does not exist (it diverges).
> So we approximate the ODE for very small r by keeping only the dominant term sr^{α}, solving this simplified equation, and showing that r(t) increases away from 0.
> This proves that $r = 0$ is an unstable equilibrium.

Separate variables:
$r^{(-\alpha)}\, dr = s\, dt$
Integrating both sides:
$\int r^{(-\alpha)}\, dr = \int s\, dt$
Left side is $r^{(1-\alpha)}/(1-\alpha)$ and right side is $s\, t + C$.

Thus
$r^{(1-\alpha)}/(1-\alpha) = s\, t + C$

Use initial condition $r(0) = r0$ to get $C = r0^{(1-\alpha)}/(1-\alpha)$. Multiply both sides by $(1-\alpha)$:
$r^{(1-\alpha)} = (1-\alpha)\, s\, t + r0^{(1-\alpha)}$

Inverting it to obtain r(t):
$r(t) = [\, (1-\alpha)\, s\, t + r0^{(1-\alpha)}\,]^{(1/(1-\alpha))}$
As t increases, the bracket grows linearly in t and r(t) increases, so small positive r_0 leads to r growing away from zero.

Therefore $r = 0$ is unstable (a repeller from the right). Any small positive perturbation moves the system away from zero and ultimately toward the interior steady state r^*.

Sign analysis summarizing dynamics
For $r > 0$ evaluating sign of $\dot{r} = s\, r^{\alpha} - A\, r = r\, [\, s\, r^{(\alpha-1)} - A\,]$.
If $0 < r < r^*$ then $s\, r^{(\alpha-1)} > s\,(r^*)^{(\alpha-1)} = A$?

Because $\alpha - 1 < 0$, $r^{(\alpha-1)}$ is decreasing in r.

So for $r < r^*$ we have $r^{(\alpha-1)} > (r^*)^{(\alpha-1)}$.

Multiply by s: $s\, r^{(\alpha-1)} > s\,(r^*)^{(\alpha-1)} = A$.

Hence $s\, r^{(\alpha-1)} - A > 0$ and $\dot{r} > 0$.

So r increases when $r < r^*$.
If $r > r^*$ then $r^{(\alpha-1)} < (r^*)^{(\alpha-1)}$ and $s\, r^{(\alpha-1)} - A < 0$ and $\dot{r} < 0$.

So r decreases when $r > r^*$.
Thus r^* is the unique attractor for all $r > 0$ and $r = 0$ repels positive initial conditions.

Trajectories starting at $r = 0$ stay there, but any infinitesimal positive perturbation sends the trajectory upward toward r^*.
The model has two equilibria: $r = 0$ and $r = r^* = [\, s / (n + a + \delta)\,]^{(1/(1-\alpha))}$.

The trivial equilibrium $r = 0$ is unstable because the nonlinear term $s\, r^{\alpha}$ dominates near zero and forces positive small r to grow; linearization is not valid at zero because $g'(0)$ is singular, but solving the dominant separable ODE $\dot{r} \approx s\, r^{\alpha}$ shows r moves away from zero. The positive equilibrium r^* is asymptotically stable; using the substitution $z = r^{(1-\alpha)}$ one obtains the exact general solution $z(t) = s/(n + a + \delta) + [\, r0^{(1-\alpha)} - s/(n + a + \delta)\,]\, e^{\{-L\, t\}}$ with $L = (1 - \alpha)(n + a + \delta)$, which implies $r(t) \to r^*$ for any $r0 > 0$ and small deviations decay exponentially at rate L. The phase-line sign analysis confirms monotone convergence to r^* from both sides within $r > 0$.

Stability conditions

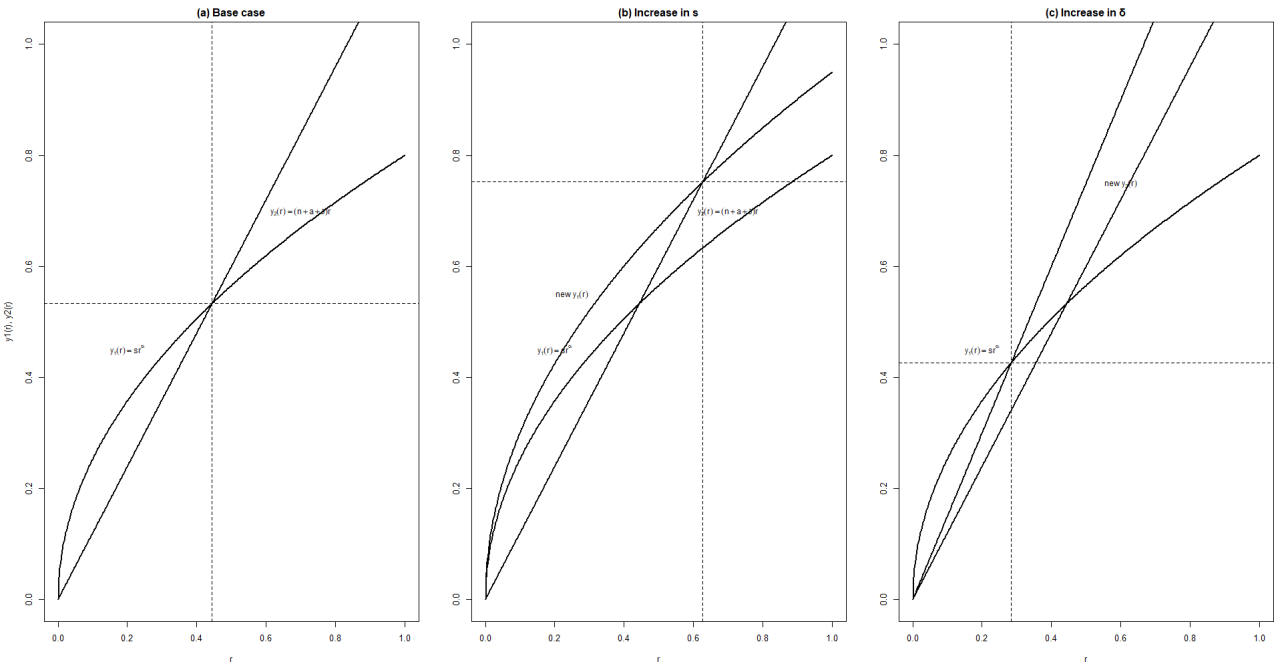
Assume $s > 0$, $0 < \alpha < 1$, $n \geq 0$, $a \geq 0$, $\delta \geq 0$ and $A = n + a + \delta > 0$.

Fixed point	Expression	Linearization / derivative g'(r)	Stability condition (sign of g')	Intuition / behaviour	Convergence rate and notes
Trivial fixed point	$r = 0$	$g'(r) = s\, \alpha\, r^{(\alpha-1)} - A$. At $r = 0$ the term $r^{(\alpha-1)}$ is singular ($r^{(\alpha-1)} \to +\infty$ as $r \to 0^+$), so $g'(0)$ is effectively $+\infty$.	Unstable (repelling). Linearization is not valid at $r = 0$ because $g'(0)$ is singular; still, the local dynamics satisfy $\dot{r} \approx s\, r^{\alpha}$ for very small r, which gives growth away from zero.	Marginal product of capital is unbounded as $r \to 0^+$, so any small positive capital stock grows. $r = 0$ is an unstable equilibrium but is absorbing only if r is exactly zero.	For very small positive r, approximate solution $r(t) = [\, (1-\alpha)\, s\, t + r0^{(1-\alpha)}\,]^{(1/(1-\alpha))}$, which increases away from zero. No exponential linear rate applies because linearization fails.

Positive steady state	r = r* = [s / (n + a + δ)]^(1/(1−α)) (exists and is positive when s > 0 and A > 0)	g'(r*) = s α (r*)^(α−1) − A. Use equilibrium relation s (r*)^(α−1) = A to get g'(r*) = (α − 1) A = −(1−α) A.	Locally asymptotically stable because g'(r*) < 0 (since 0 < α < 1 and A > 0).	Investment per effective worker equals dilution at r*. Diminishing returns make investment increase slower than dilution beyond r*, so deviations decay back to r*. This equilibrium attracts all positive initial r (global attractor on r > 0).	Convergence is exponential in the transformed variable z = r^(1−α). Rate in z is L = (1−α) A. Small deviations e = r − r* obey e˙ ≈ g'(r*) e = −L e, so e(t) ≈ e0 e^(−L t). The full solution gives r(t) = [z_p + (r0^(1−α) − z_p) e^(−L t)]^(1/(1−α)), where z_p = s / A.

- Existence of r*: requires s > 0 and A = n + a + δ > 0.
- Basin of attraction: all initial r0 > 0 converge to r*. The point r = 0 is repelling for any positive perturbation.
- Rate interpretation: larger A (faster population, tech growth, or depreciation) or larger (1−α) imply faster convergence in z; α near 1 slows convergence.

Figure 9.1 simulation



Graphical stability analysis of the Solow–Swan model

In the Solow model, the law of motion for capital per effective labour is:

$$\dot{r} = sr^{\alpha} - (n + a + \delta)r$$

Defining the two components:

- $y_1(r) = sr^{\alpha}$— *effective investment per worker*
- $y_2(r) = (n + a + \delta)r$— *break-even investment*
  (the amount of capital needed to keep r constant)

These two curves determine how $r(t)$evolves.

The model behaves depending on whether actual investment is above or below break-even investment.

Panel (a): Base Case

Parameters:

- $s = 0.8$
- $\alpha = 0.5$
- $n = 0.3$, $a = 0.3$, $\delta = 0.4$

Shape of the curves

- $y_1 = sr^{\alpha}$is concave, starting steep but flattening as $r$increases.
- $y_2 = (n + a + \delta)r$is a straight upward sloping line.

Two intersections

The curves intersect at:

- r_1^*(near 0)
- r_2^*(the economically relevant steady state)

Stability analysis

From the figure:

- For $0 < r < r_1^*$:
  $y_1 > y_2 \Rightarrow \dot{r} > 0 \Rightarrow$ capital grows
  → $r_1^*$is unstable (if you move away, you never return)
- For $r > r_1^*$but $r < r_2^*$:
  $y_1 > y_2 \Rightarrow$ capital grows ⇒ moves toward r_2^*
- For $r > r_2^*$:
  $y_1 < y_2 \Rightarrow \dot{r} < 0 \Rightarrow$ capital declines
  → $r_2^*$is stable

Economic intuition

- Very poor economies (tiny r) grow because marginal product of capital is extremely high.
- Economies above the steady state shrink because they cannot maintain their capital stock.

- The economy always converges to the higher steady state r_2^*.

Panel (b): Increase in Savings Rate s

We increase s, which shifts the investment curve $y_1(r) = sr^\alpha$upward.

What changes

- $y_1(r)$becomes higher at every r
- It intersects the break-even line at a higher r_2^*
- Steady state shifts rightward
- The new curve lies above the old y_1— shown in your graph

Economic intuition

- Higher savings → more investment → higher capital accumulation
- Steady-state capital per worker increases
- Long-run output and consumption *may* rise (unless overshooting golden rule)

Panel (c): Increase in Depreciation δ

(Or alternately increase in n)

An increase in δor $n$shifts $y_2(r)$upward, because:

$$y_2 = (n + a + \delta)r$$

What changes

- Break-even investment line becomes *steeper*
- It intersects the unchanged investment curve at a lower r_2^*
- The new line lies above the old one — exactly as in your figure

Economic intuition

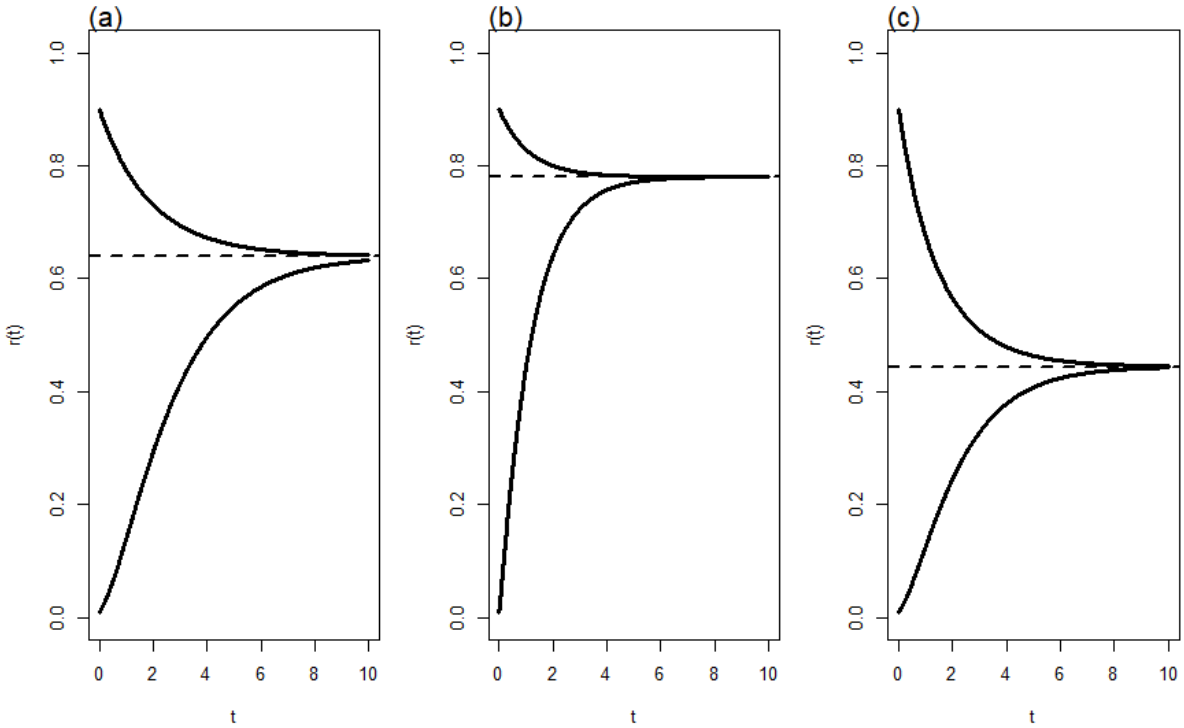
High depreciation or population growth means more capital must be replaced each year.
Thus:

- Harder to maintain capital
- Steady state drops
- Long-run output per worker falls

Figure 9.1 is the *static* graphical comparison of investment vs. break-even lines.
Figure 9.2 is the *dynamic* time-series simulation showing the solution trajectories of:

$$\dot{r} = sr^\alpha - (n + a + \delta)r$$

Figure 9.2 simulation



Panel (a): Baseline dynamics

Parameters:
s = 0.8, α = 0.5, n = 0.3, a = 0.3, δ = 0.4

What happens in the plot

- Two economies start with very different initial capital–labour ratios (r(0)=0.01 and r(0)=0.9).
- Both converge monotonically toward the same steady state.
- The upper curve (high r0) falls downward.
- The lower curve (low r0) rises upward.
- They meet exactly at the steady-state r*.

Economic intuition

- With α = 0.5, diminishing returns to capital are moderate.
- A low-capital economy has a high marginal product of capital, so it grows rapidly.
- A high-capital economy faces excess depreciation + dilution relative to output, so r(t) falls.
- Ultimately, both economies settle where investment per effective worker equals depreciation + dilution:

$$s\, r^{\alpha} = (n + a + \delta)\, r.$$

This is the standard Solow convergence result.

Panel (b): Lower capital share (α = 0.1)

Parameters:
$s = 0.8$, $\alpha = 0.1$, $n = 0.3$, $a = 0.3$, $\delta = 0.4$

What happens in the plot

• The steady state is higher than in panel (a).
• The upper trajectory (starting at $r(0)=0.9$) now converges from above, but only slightly — it stays just above the steady-state before gently approaching it.
• The lower trajectory grows very quickly at first and then flattens.

Why this happens mathematically

Lower α strengthens diminishing returns:
• When α is small, the output response to an increase in capital is very weak.
• This means depreciation + dilution grows *faster* than additional output at high capital levels.
• As a result, the steady state shifts upward (because $s/(n+a+\delta)$ stays the same, but the transformation $r = z^{(1/(1-\alpha))}$ amplifies steady state levels).

Economic intuition

• With **$\alpha = 0.1$**, capital is far less productive.
• Poor economies still grow quickly at the beginning — because MPK is still very high at low r — but they hit diminishing returns *much sooner*.
• Rich economies (starting at high r) lose capital quickly at first because the small α means their production is not enough to offset $n + a + \delta$.
• The steady state is reached, but with much slower adjustment near r^*.

Lower α makes the production function flatter. This increases the gap between marginal product at low r and marginal product at high r, slowing convergence near the steady state.

Panel (c): Higher population growth (n = 0.5)

Parameters:
$s = 0.8$, $\alpha = 0.5$, $n = 0.5$, $a = 0.3$, $\delta = 0.4$

What happens in the plot

• The steady state is lower than in panel (a).
• The upper curve falls much faster because higher n increases dilution.
• The lower curve grows, but more slowly, because more output must go toward capital widening.

Economic intuition

Higher population growth means the capital stock must be spread across more effective workers. This increases the required break-even investment:

$$\text{Break-even} = (n + a + \delta)\, r$$

When n increases:
• break-even line becomes steeper
• steady-state r^* becomes smaller
• the economy must accumulate capital more slowly to balance out higher dilution
• high-r economies shrink faster because depreciation + dilution dominates output

In words:

A higher population growth rate makes it harder to maintain capital per worker, so the steady state drops and convergence happens toward a lower r*

Panel	Changed Parameter	Steady State Effect	Dynamic Effect	Economic Interpretation
(a) Baseline	α = 0.5, n = 0.3	Normal r*	Symmetric convergence from above and below	Standard Solow convergence driven by diminishing returns
(b) α ↓ (0.2)	Lower α	r* ↑	Convergence slower near r*, upper curve flattens	Capital is less productive; diminishing returns stronger
(c) n ↑ (0.5)	Higher n	r* ↓	High-r economies fall faster, low-r economies grow slower	More dilution makes it harder to maintain r

These two figures together illustrate the central policy implication of the Solow model: long-run living standards depend only on parameters such as saving, population growth, and technology, not on initial conditions. Figure 9.1 shows visually how policy variables like *s* or *n* shift the steady state r_2*. Figure 9.2 confirms that regardless of where an economy starts, it will converge toward the new r_2*, though at different speeds. This joint graphical-simulation perspective helps policymakers understand how changes in saving incentives, education-driven labor efficiency, demographic policies, or depreciation-related investment incentives can permanently raise or lower the long-run capital stock and output per worker. It also highlights why poor countries can grow faster (conditional convergence), and why sustained growth ultimately requires technological progress.

DDD_ASSSIGNMENT_6

Michelle

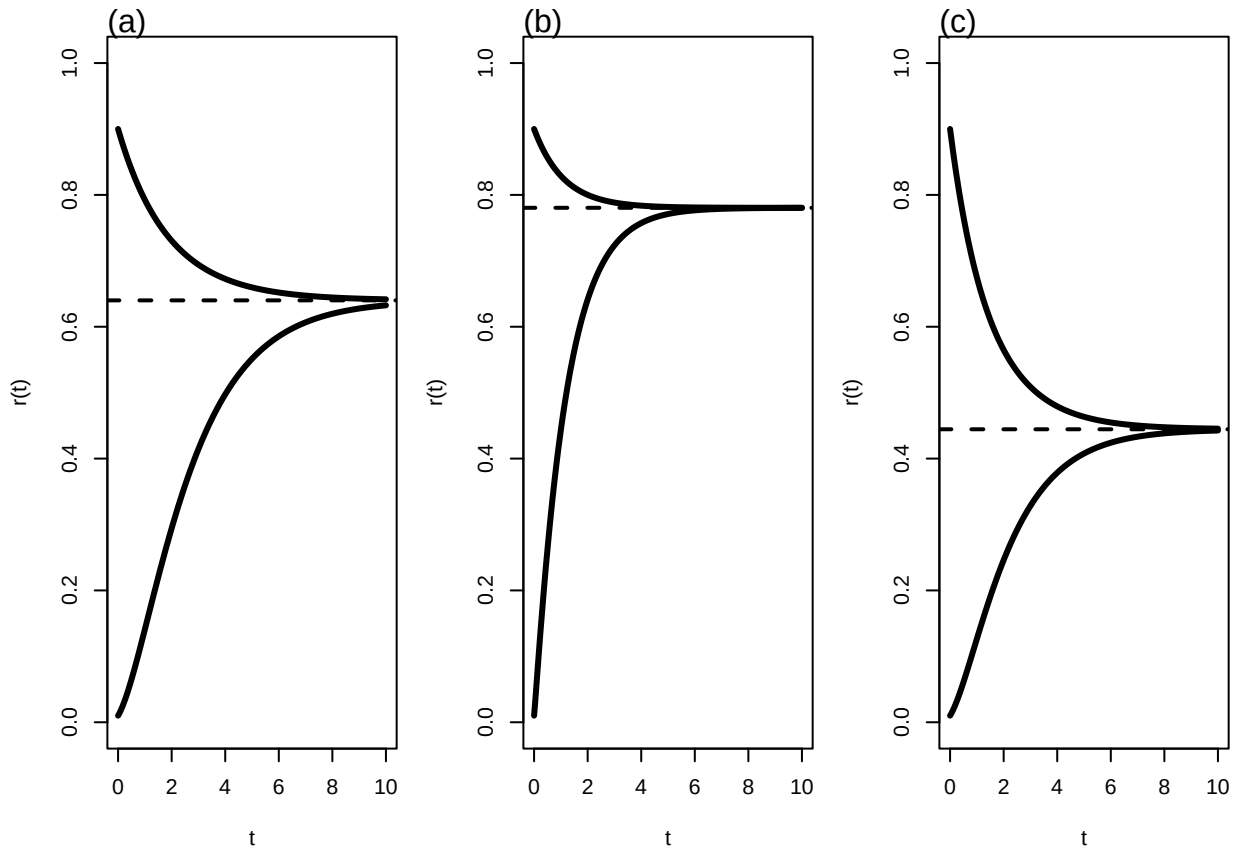
2025-11-19

```r
### --- Solow Model Simulation

library(deSolve)
```

```
## Warning: package 'deSolve' was built under R version 4.3.3
```

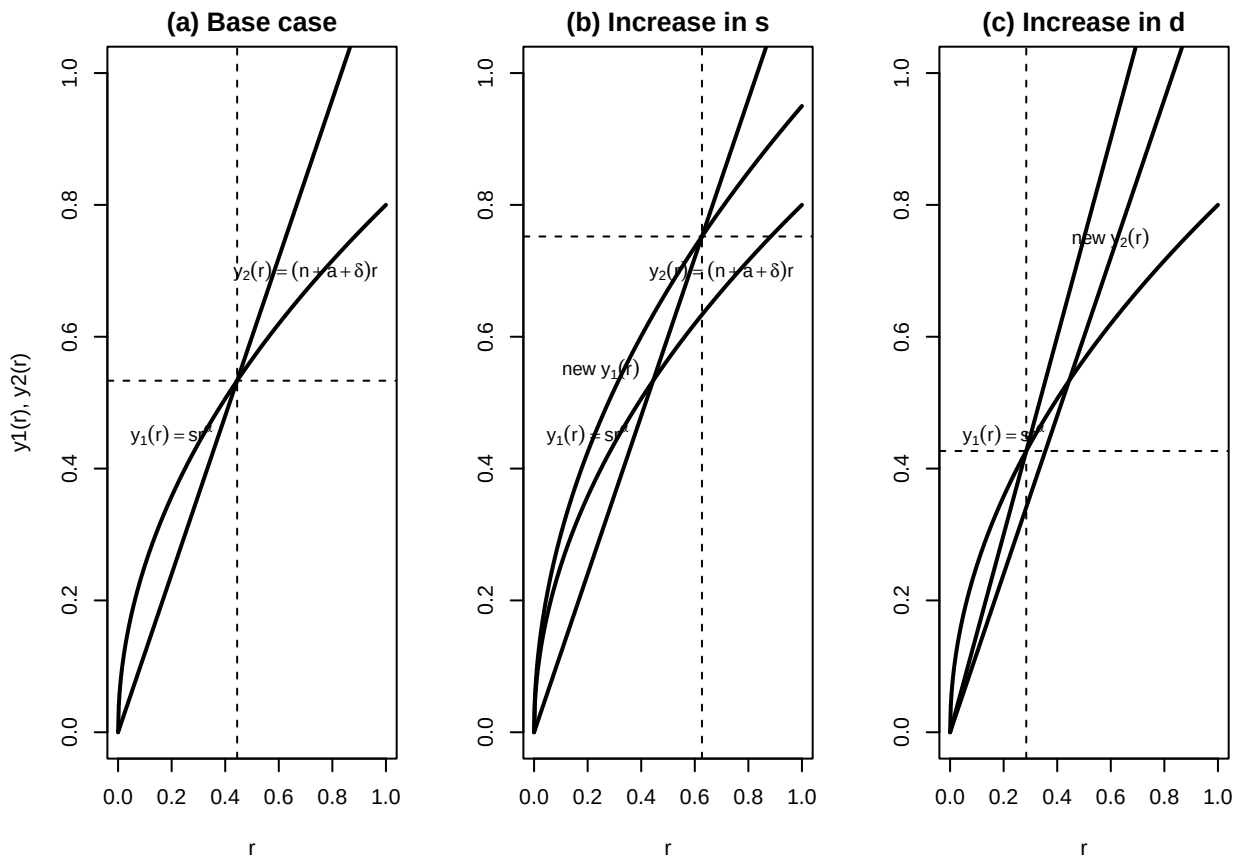
```r
# Solow ODE function
solow_system <- function(t, r, parms){
  with(as.list(parms), {
    dr <- s * r^alpha - (n + a + delta) * r
    list(dr)
  })
}

# Function to simulate r(t) for given parameters
simulate_solow <- function(s, alpha, n, a, delta, r0, t_end=10){
  parms <- list(s=s, alpha=alpha, n=n, a=a, delta=delta)
  times <- seq(0, t_end, by=0.01)
  out <- ode(y=c(r=r0), times=times, func=solow_system, parms=parms)
  as.data.frame(out)
}

# ---- Parameters (baseline) ----
s <- 0.8
alpha_base <- 0.5
n_base <- 0.3
a <- 0.3
delta <- 0.4

# Initial conditions
r0_vals <- c(0.01, 0.9)


par(mfrow=c(1,3), mar=c(4,4,2,1))

### -------- PANEL (a): Base case -------- ###
alpha <- alpha_base
n <- n_base
r_eq <- (s/(n+a+delta))^(1/(1-alpha))

plot(NULL, xlim=c(0,10), ylim=c(0,1), xlab="t", ylab="r(t)")
```

```r
for(r0 in r0_vals){
  out <- simulate_solow(s, alpha, n, a, delta, r0)
  lines(out$time, out$r, lwd=3)
}
abline(h=r_eq, lty=2, lwd=2)
mtext("(a)", adj=0)

### -------- PANEL (b): changing alpha to 0.1 (different from txt)-------- ###
alpha <- 0.1
n <- n_base
r_eq <- (s/(n+a+delta))^(1/(1-alpha))

plot(NULL, xlim=c(0,10), ylim=c(0,1), xlab="t", ylab="r(t)")
for(r0 in r0_vals){
  out <- simulate_solow(s, alpha, n, a, delta, r0)
  lines(out$time, out$r, lwd=3)
}
abline(h=r_eq, lty=2, lwd=2)
mtext("(b)", adj=0)

### -------- PANEL (c): Increase n to 0.5 -------- ###
alpha <- alpha_base
n <- 0.5
r_eq <- (s/(n+a+delta))^(1/(1-alpha))

plot(NULL, xlim=c(0,10), ylim=c(0,1), xlab="t", ylab="r(t)")
for(r0 in r0_vals){
  out <- simulate_solow(s, alpha, n, a, delta, r0)
  lines(out$time, out$r, lwd=3)
}
abline(h=r_eq, lty=2, lwd=2)
mtext("(c)", adj=0)
```




```r
# Figure 9.1 - Graphical stability analysis (Panels a, b, c)


# Parameters for panel (a)
s_base <- 0.8
alpha_base <- 0.5
n_base <- 0.5
a_base <- 0.3
delta_base <- 0.4

# Domain for r
r <- seq(0, 1, length.out = 400)

# Functions
y1 <- function(r, s, alpha) s * r^alpha
y2 <- function(r, n, a, delta) (n + a + delta) * r

# Steady state function
r_star <- function(s, alpha, n, a, delta) {
  (s / (n + a + delta))^(1 / (1 - alpha))
}

# PANEL (a) base case
par(mfrow = c(1,3), mar = c(4,4,2,1))

r_eq_a <- r_star(s_base, alpha_base, n_base, a_base, delta_base)
```

```r
plot(r, y1(r, s_base, alpha_base), type = "l", lwd = 2,
     xlab = "r", ylab = "y1(r), y2(r)", ylim = c(0,1), xlim = c(0,1))
lines(r, y2(r, n_base, a_base, delta_base), lwd = 2)

# Steady-state dashed lines
abline(v = r_eq_a, lty = 2)
abline(h = y1(r_eq_a, s_base, alpha_base), lty = 2)

# Labels
text(0.2, 0.45, expression(y[1](r)==s*r^alpha), cex = 0.9)
text(0.7, 0.7, expression(y[2](r)==(n+a+delta)*r), cex = 0.9)
title("(a) Base case")

# PANEL (b) - higher saving rate
s_high <- 0.95
r_eq_b <- r_star(s_high, alpha_base, n_base, a_base, delta_base)

plot(r, y1(r, s_base, alpha_base), type = "l", lwd = 2,
     xlab = "r", ylab = "", ylim = c(0,1), xlim = c(0,1))
lines(r, y2(r, n_base, a_base, delta_base), lwd = 2)

# New investment curve (solid black overlay)
lines(r, y1(r, s_high, alpha_base), lwd = 2)

# Steady state new level
abline(v = r_eq_b, lty = 2)
abline(h = y1(r_eq_b, s_high, alpha_base), lty = 2)

text(0.25, 0.55, expression(paste("new ", y[1](r))), cex = 0.9)
text(0.2, 0.45, expression(y[1](r)==s*r^alpha), cex = 0.9)
text(0.7, 0.7, expression(y[2](r)==(n+a+delta)*r), cex = 0.9)
title("(b) Increase in s")

# PANEL (c) - higher depreciation
delta_high <- 0.7
r_eq_c <- r_star(s_base, alpha_base, n_base, a_base, delta_high)

plot(r, y1(r, s_base, alpha_base), type = "l", lwd = 2,
     xlab = "r", ylab = "", ylim = c(0,1), xlim = c(0,1))
lines(r, y2(r, n_base, a_base, delta_base), lwd = 2)

# New y2 curve with higher depreciation
lines(r, y2(r, n_base, a_base, delta_high), lwd = 2)

# New steady state lines
abline(v = r_eq_c, lty = 2)
abline(h = y1(r_eq_c, s_base, alpha_base), lty = 2)

text(0.2, 0.45, expression(y[1](r)==s*r^alpha), cex = 0.9)
text(0.6, 0.75, expression(paste("new ", y[2](r))), cex = 0.9)
title("(c) Increase in  ")
```


(a) Base case
(b) Increase in s
(c) Increase in d
y1(r), y2(r)
r
$y_2(r) = (n+a+\delta)r$
$y_1(r) = sr^{\alpha}$
new $y_1(r)$
new $y_2(r)$
0.0
0.2
0.4
0.6
0.8
1.0